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“Report on User Interaction Controls for the 3D Solar System Simulation”

Introduction

This report outlines the user interaction controls implemented in the 3D Solar System simulation project. The primary focus is on detailing the mouse and keyboard controls that facilitate navigation and interaction within the virtual environment. Effective user controls are crucial for an immersive and intuitive experience, enabling users to explore the solar system model seamlessly.

Control Scheme Overview

The simulation incorporates a combination of mouse and keyboard inputs to allow users to manipulate the camera view, select celestial bodies, and control the movement of planets. The controls are designed to be intuitive, leveraging common interaction paradigms to enhance usability.

Mouse Controls

The mouse interface provides mechanisms for rotating the camera view and zooming in and out of the scene. The specific mouse interactions are as follows:

1. **Left Mouse Button (LMB) + Drag: Rotate Camera**

- **Functionality:** Allows the user to rotate the camera around the target point, providing different perspectives of the solar system.
- **Implementation Details:**
  - **Activation:** Press and hold the left mouse button while moving the mouse.
  - **Behavior:** Horizontal mouse movement adjusts the camera's yaw (left-right rotation), while vertical movement adjusts the camera's pitch (up-down rotation).
  - **Constraints:** The pitch is clamped between -89° and +89° to prevent the camera from flipping over.

2. **Mouse Wheel: Zoom In/Out**

- **Functionality:** Enables the user to zoom the camera closer to or farther from the target object or the overall scene.

- **Implementation Details:**
  - **Zoom In (Scroll Up):**
  - **Action:** Scroll the mouse wheel upward.

- **Effect:** Decreases the camera distance (`cameraDistance`) by 0.5 units per scroll tick.
- **Constraints:**
- **General Zoom:** Minimum distance is clamped to 2.0 units to prevent the camera from getting too close to the scene.
- **Held Object Zoom:** When a specific celestial body is selected (`heldObject`), the minimum distance is dynamically set based on the selected object's radius to avoid clipping.

- **Zoom Out (Scroll Down):**
- **Action:** Scroll the mouse wheel downward.
- **Effect:** Increases the camera distance (`cameraDistance`) by 0.5 units per scroll tick.
- **Constraints:** The maximum zoom distance is clamped to 100.0 units to prevent the camera from moving too far away from the scene.

Keyboard Controls

The keyboard interface provides controls for selecting celestial bodies, controlling planetary movement, and exiting the simulation. The specific keyboard interactions are as follows:

1. **Numeric Keys '0' to '8': Select Sun and Planets**

   - **Functionality:** Allows the user to select a specific celestial body (Sun or any of the eight planets) to focus the camera on.
   - **Implementation Details:**
   - **Keys and Corresponding Bodies:**
   - **'0':** Sun
   - **'1':** Mercury
   - **'2':** Venus
   - **'3':** Earth
   - **'4':** Mars
   - **'5':** Jupiter
   - **'6':** Saturn
   - **'7':** Uranus
   - **'8':** Neptune
   - **Behavior:** Pressing a key from '0' to '8' sets the `heldObject` to the corresponding celestial body, causing the camera to zoom in on the selected object.
   - **Feedback:** The selected object's name is displayed on the screen to confirm selection.
   - **Release Behavior:** Releasing the key deselects the object, resetting the camera focus to the default target point (typically the Sun).

2. **Keys '+' and '-': Control Planetary Movement**

   - **Functionality:** Enables the user to control the rotational movement of the planets around the Sun.
   - **Implementation Details:**
   - **'+' Key:**

- **Action:** Increments the `orbitAngle` by 1 degree, causing all planets to rotate anti-clockwise around the Sun.
- **Additional Behavior:** Simultaneously rotates the Sun on its own axis based on the defined `planetRotationSpeeds`.
- **'-' Key:**
- **Action:** Decrements the `orbitAngle` by 1 degree, causing all planets to rotate clockwise around the Sun.
- **Additional Behavior:** Simultaneously rotates the Sun on its own axis in the opposite direction.
- **Constraints:** The `orbitAngle` and `planetRotationAngles` are maintained within the range of [0°, 360°) to ensure continuous rotation without overflow.

3. **ESC Key: Exit Program**

- **Functionality:** Allows the user to gracefully exit the simulation.
- **Implementation Details:**
- **Action:** Pressing the ESC key triggers the `cleanup` function, which deletes all loaded textures and performs necessary resource deallocation before terminating the program.
- **Feedback:** A console message "ESC key pressed. Exiting the program." is displayed to confirm the exit action.

Combined Mouse and Keyboard Interactions

The simulation supports combined interactions where mouse movements and keyboard inputs can be used simultaneously to navigate and interact with the solar system model effectively. For example, a user can select a planet using the numeric keys and then use the mouse to rotate the camera around the selected planet, providing a detailed view.

Detailed Control Implementation

Mouse Button Functionality

1. **Left Mouse Button (LMB)**

- **Usage:** Initiates camera rotation.
- **Behavior:**
- **On Press (State = GLUT_DOWN):**
- Sets `isDragging` to `true`.
- Records the initial mouse position (`lastMouseX`, `lastMouseY`) to calculate movement deltas.
- **On Release (State = GLUT_UP):**
- Sets `isDragging` to `false`, stopping the camera rotation.
- **Drag Movement:**
- While `isDragging` is `true`, mouse movements adjust the `cameraYaw` and `cameraPitch` based on the movement deltas, enabling smooth camera rotation.
2. **Mouse Wheel**

- **Usage:** Controls zooming functionality.
- **Behavior:**

- **Scroll Up (Button = 3, State = GLUT_DOWN):**
- Decreases `cameraDistance` by 0.5 units per scroll tick.
- Applies dynamic minimum distance constraints based on whether an object is held.
- **Scroll Down (Button = 4, State = GLUT_DOWN):**
- Increases `cameraDistance` by 0.5 units per scroll tick.
- Applies a maximum zoom-out limit to maintain scene visibility.

Keyboard Button Functionality

1. **Numeric Keys ('0' - '8')**

   - **Usage:** Selects specific celestial bodies (Sun and planets) to focus the camera on.
   - **Behavior:**
   - **On Press:**
   - Sets `heldObject` to the corresponding index (0 for Sun, 1-8 for Mercury to Neptune).
   - Outputs a console message indicating the selected object.
   - **On Release:**
   - Resets `heldObject` to `-1`, deselecting the object and reverting camera focus.
2. **'+' and '-' Keys**

   - **Usage:** Controls the orbital movement of planets.
   - **Behavior:**
   - **'+' Key Press:**
   - Increments `orbitAngle` by 1 degree, facilitating anti-clockwise rotation.
   - Simultaneously increments the Sun's rotation angle (`planetRotationAngles[0]`) based on predefined rotation speeds.
   - **'-' Key Press:**
   - Decrements `orbitAngle` by 1 degree, facilitating clockwise rotation.
   - Simultaneously decrements the Sun's rotation angle (`planetRotationAngles[0]`) based on predefined rotation speeds.
   - **Angle Maintenance:**
   - Ensures `orbitAngle` and `planetRotationAngles[0]` remain within [0°, 360°) to sustain continuous rotation.
3. **ESC Key**

   - **Usage:** Exits the simulation.
   - **Behavior:**
   - **On Press:**
   - Calls the `cleanup` function to delete textures and free resources.
   - Terminates the program gracefully with an exit status of `0`.

User Feedback Mechanisms

To enhance user experience and provide clear feedback on interactions, the simulation incorporates on-screen instructions and console messages:

- **On-Screen Instructions:**

- Displayed in the upper-left corner of the window, providing a quick reference to available controls.
- Includes details on mouse and keyboard interactions, ensuring users can easily navigate and utilize the simulation's features.

- **Console Messages:**

- Provide real-time feedback on actions such as selecting or deselecting celestial bodies.
- Inform users of significant events like exiting the program, aiding in debugging and user awareness.

Conclusion

The mouse and keyboard controls in the 3D Solar System simulation make it easy and enjoyable to use. Users can smoothly move the camera, choose different planets or the Sun, and control how the planets move. By dragging the mouse, you can rotate the view, using the mouse wheel lets you zoom in and out, and the keyboard enables us to select objects and manage their movements. This combination of controls makes exploring the solar system fun and flexible. In the future, we could add features like right-click options, touch gestures for devices with touchscreens, and the ability to customize keys to make the simulation easier to use and more accessible for everyone.